# Autumn Semester Examination 2024

Subject: Mathematical Physics II
Full Marks: 60

Subject Code: PHYUGMCC2306
Time: 2 hrs 30 minutes

*[Each question carries 10 marks, Answer any six questions]*

1. The Chebyshev differential equation is given as $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + n^2y = 0$, where $n$ is an arbitrary integer.

   (a) Test the singularity of the differential equation at $x = 0, \pm 1$.

   (b) Considering the trial solution as $y = \sum_{k=0}^{\infty} a_k x^{p+k}$, obtain the indicial equation and find the possible values of $p$.

   (c) Derive the recurrence relation between the power series coefficients ($a_k$'s) and find the general solution of the differential equation. (3+2+5)

2. (a) Define periodic function.

   (b) State Dirichlet's condition and define the Fourier series of a periodic function $f(x)$ with period $2\pi$.

   (c) For an odd function, derive why its Fourier series contains only sine terms.

   (d) Find the Fourier series for $f(x) = x^2$, in $-\pi < x < \pi$. (1+3+2+4)

3. Consider the differential equation $xy'' + y' + \left(\frac{a}{x} + b + cx + dx^2\right) y = 0$ where, $a = -\frac{m^2}{4}$ $(m > 0)$; $b = \alpha$ $(\alpha > 0)$; $c = \frac{\epsilon}{2}$ $(\epsilon > 0)$ and $d = -\frac{F}{4}$ $(F > 0)$.

   (a) Test the singularity of the differential equation at $x = 0$.

   (b) Considering the Fuch theorem, comment on the results obtained in (a)

   (c) Obtain the indicial equation using a suitable ascending trial power series around $x = 0$.

   (d) Using the larger root of the indicial equation, show that the solution can be written as

$$y(x) = a_0 x^{\frac{m}{2}} \left[1 - \frac{\alpha}{m+1}x + \left\{\frac{\alpha^2}{2(m+1)(m+2)} - \frac{E}{4(m+2)}\right\} x^2 + ...\right]$$

(2+1+3+4)

4. (a) Define the Gamma function $\Gamma(x)$ and explain its relation to factorials.

   (b) Show that (i) $\Gamma(n+1) = n!$ for positive integers $n$. (ii) $\beta(m,n) = \beta(n,m)$ (iii) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ (2+3+3+2)

5. (a) Describe the Parseval's identity.

   (b) A square wave function is defined as,

$$f(x) = \begin{cases} +1 & \text{for } 0 < x < \pi \\ -1 & \text{for } -\pi < x < 0 \end{cases}$$

   Find its Fourier series representation.

   (c) Using the findings of (b), prove that $\frac{\pi^2}{6} = \sum_1^{\infty} \frac{1}{n^2}$ (2+3+5)

6. Using the generating function of the Bessel's polynomial, show that

   (a) $J_{3/2} = \sqrt{\frac{2}{\pi x}} \left[\frac{\sin x}{x} - \cos x\right]$

   (b) $\cos(x \sin\varphi) = J_0(x) + 2\sum_{k=1}^{\infty} J_{2k}(x) \cos 2k\varphi$;  $\sin(x\sin\varphi) = 2\sum_{k=0}^{\infty} J_{2k+1}(x) \sin(2k+1)\varphi$

(4+6)

7. Adopting the separation of variable technique, find the general solution of the three-dimensional diffusion equation (in Cartesian coordinates) defined as,

$$\nabla^2 u = \frac{1}{h^2}\frac{\partial u}{\partial t}$$

where, $h$ is the diffusion constant. Comment on your result. Describe an example in which you can realize the diffusion equation in one dimension. (7+1+2)

8. Using the definitions of the Legendre polynomial $P_l(x)$ and Hypergeometric function $F(a, b, c; x)$, show that

(a) $x^2 = \frac{1}{3}[2P_2(x) + P_0(x)]$; $x^3 = \frac{1}{5}[2P_3(x) + 3P_1(x)]$

(b) $(1+x)^n = F(-n, 1, 1; -x)$; $\log(1+x) = xF(1, 1, 2; -x)$

[(2.5 + 2.5) + (2.5 + 2.5)]

9. If $L_n(x)$ is the nth order Laguerre polynomial, then prove that

$$\frac{e^{-\frac{xt}{1-t}}}{1-t} = \sum_{n=0}^{\infty} L_n(x)t^n$$

Using this relation, show that

# Autumn(Odd) Semester Examination - 2024

**Course Code:** PHYUGMCC2305
**Course Title:** Thermal Physics
**Department of Appearing Students:** Physics, SEM-III, 2$^{nd}$Year
**Full Marks:** 60 **Time:** 2hr 30 min

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[Use separate answer sheet for Group-A and Group-B]

## Group-A

$6 \times 5 = 30$

Answer any **five** questions. Each question carries 6 marks.

1. Energy distribution law for a system of an ideal gas molecule at temperature T is given by

$$N(E)dE = \frac{2\pi N}{(\pi kT)^{3/2}} \exp(-\frac{E}{kT})\sqrt{E}dE,$$

where $N$ is the total number of molecules. Calculate the (i) average and (ii) most probable energy. [3 +3 =6]

2. According to Maxwell-Boltzmann distribution in one direction, the probability density that a molecule has a velocity of v, is given by

$$f(v) = Ae^{-\frac{Mv^2}{2RT}}.$$

Find the value of constant A and also find the average and most probable velocity of the particles following the above distribution. [2+2+2=6]

3. How the total number of degrees of freedom is related to the ratio of specific heat of the system. Using laws of equipartition of energy, obtain the average energy per molecule in the following cases: (i) a gas of diatomic molecules (two atoms are connected by a rigid weightless rod), (ii) Tri-atomic linear molecules (Three atoms are connected by two rigid weightless rods). [3+1.5+1.5=6]

4. (a) At what temperature is the mean translational K.E. of a molecule equal to that of a singly charged ion of the same mass that has been accelerated from rest through a potential difference of 1 volt? Neglect relativistic effect.

   (b) A narrow molecular beam makes its way into a vessel filled with gas under low pressure. Find the mean free path of the molecules if the beam intensity decreased to 1/n times its initial intensity over a distance $\Delta l$. [3+3=6]

5. (a) Write down the van der Waals equation of state for a real gas and explain the significance of correction terms.

(b) Show that with the increase of volume per mole, the van der Waals equation of state tends to that of an ideal gas. [(1+3)+2=6]

6. Assuming that the critical state corresponds to the point of inflexion of an isothermal, find the critical constants ($P_c, V_c$ and $Tc$)) and the critical coefficients for a gas obeying the following equation of state

$$p = \frac{RT}{V-b} - \frac{a}{TV^2},$$

where $a$ and $b$ are constants. [4.5+1.5 =6]

7. (a) 'Brownian motion may be considered as analogous to the heat motion of the molecules but on a much-reduced scale'- explain the significance of the statement.

(b) Write down the expression (no derivation) for the coefficient of viscosity and explain how the coefficient of viscosity ($\eta$) varies with Pressure and temperature. [2+2+2=6]

## Group-B

$10 \times 3 = 30$

Answer any **three** questions. Each question carries 10 marks.

8. (a) What are extensive and intensive variables? Identify extensive and intensive variables in the given list: volume (V), length (L), area (A), charge (Q), magnetization (I), pressure (P), tension (F), surface tension (S), EMF (E), magnetic intensity (H), density ($\rho$).

(b) Using concept of the first law of thermodynamics show that for an adiabatic process $TV^{\gamma-1}$ = constant, where the symbols have their usual meanings.

(c) A quantity of air at 127°C and 2.0 atmospheric pressure is suddenly compressed to 40% of its original volume. Find the final pressure and temperature. [(2+2)+3+3=10]

9. (a) State the Kelvin-Planck statement of the second law of thermodynamics.

(b) With a suitable diagram explain the processes involved in a Carnot cycle. Hence obtain an expression for the efficiency of a Carnot cycle.

(c) A Carnot engine has its source at 200°C and its sink is maintained at a constant temperature by means of ice at 0°C. If it is working at the rate of 200 watts, how much ice will melt in one minute? [$L_{ice}$=80 cal/gm = $80\times10^3 \times 4.2$ J/kg] [2+5+3=10]

10. (a) State and prove Carnot's theorem.

(b) Derive Maxwell's thermodynamic relations [4+6=10]

11. (a) Derive the Clausius-Clapeyron equation: $\left(\frac{dp}{dT}\right)_{sat} = \frac{L}{T(v_{vap}-v_{liq})}$. Symbols have their usual meanings.

(b) Using the specific heat equation:

$$C_P - C_V = T\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_P$$

show that: $C_P - C_V = R$ for perfect gas obeying equation of state $PV = RT$

(c) With the help of a suitable diagram explain the working principle of an Otto cycle. [3+3+4=10]

**End-Semester Examination - 2024 (Autumn)**
**MATUGMCC2305/MATUGMIN2303**
**(Ordinary Differential Equations)**
**Semester - III (4 Year B.Sc. Honours in Mathematics)**

**Full Marks – 60** **Time – 2 Hours 30 Minutes**

**Notations and symbols have their usual meaning.**

**Section A.** Answer ALL questions. Each question carries one mark.

1. The differential equation of the two parameters family of curves $y = a + \frac{b}{x}$.
   (a) $\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$.
   (b) $\frac{d^2y}{dx^2} + \frac{2}{x}\frac{dy}{dx} = 0$.
   (c) $2x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$
   (d) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$.
2. Integrating factor of the differential $xdy - ydx$ is
   (a) $\frac{1}{x^2}$
   (b) $\frac{1}{y^2+1}$
   (c) $\frac{1}{x+y}$
   (d) $\frac{1}{xy^2}$
3. Complete primitive of the differential equation $y = px + f(p)$, where $p = \frac{dy}{dx}$ is
   (a) $y = cx + f(c)$
   (b) $y = c^2x + f(c)$
   (c) $y = cx + f(c^2)$
   (d) $y = c/x + f(1/c)$
4. The orthogonal trajectories of the family of straightlines of $y = mx$ is
   (a) $x^2 + y^2 = c$
   (b) $xy = c$
   (c) $y = x + c$
   (d) $x^2 - y^2 = c$

5. The degree of the differential equation $\frac{d^2y}{dx^2}+x^2\frac{dy}{dx}+x\sin y=0$ is

(a) 3

(b) 2

(c) 1

(d) 0

6. If the equation $Mdx+Ndy=0$ has one and only one solution, how many integrating factor exist?

(a) 1

(b) 0

(c) Infinite

(d) Finite

7. For the initial value problem $\frac{dy}{dx}+y=2x$, with $y(0)=1$, then the value of $y(2)$ is

(a) $e^{-2}$

(b) $2e$

(c) $5e^{-2}$

(d) $2+3e^{-2}$

8. Particular solution of the differential equation $(D^2-1)y=e^{2x}$, where $D\equiv\frac{d}{dx}$ is

(a) $\frac{e^{2x}}{3}$

(b) $\frac{e^{-2x}}{3}$

(c) $\frac{2x}{3}$

(d) $\frac{2x^2}{3}$

9. Which of the following is not a liner differential equation

(a) $\frac{d^2y}{dx^2}-x\frac{dy}{dx}+xy=\sin x$

(b) $e^x\frac{d^3y}{dx^3}=cosx\frac{d^2y}{dx^2}$

(c) $\frac{d^3y}{dx^3}-\frac{dy}{dx}-y=x^2\sin x$

(d) $\frac{dy}{dx}=3\cos(x+2y)$

10. The differential equation $\frac{dy}{dx}=\frac{x-2y}{3x+2y}$ is a

(a) First order homogeneous and non-linear equation

(b) First order homogeneous and linear equation

(c) First order nonhomogeneous and non-linear equation

(d) First order nonhomogeneous and linear.

**Section B.** Answer ALL questions. Each question carries one mark. $10\times 1=10$

11. Write the solution of $y = px + \frac{a}{p}$, where $p = \frac{dy}{dx}$.

12. What is the order of the differential equation whose general solution is $ax^2 + by^2 = 1$, where a and b are arbitrary constant. 2nd

13. Write down the necessary and sufficient condition for $Mdx + Ndy = 0$ to be exact .

14. Determine Wronskian value of $y_1 = x, y_2 = 1$

15. Find the general solution of $(D^2 - 1)y = 0, D = \frac{d}{dx}$

16. If the differential equation $Mdx + Ndy = 0$ is both homogeneous and exact ,write the complete primitive.

17. What is the integrating factor of the differential equation $\frac{dy}{dx} + py = Q$.

18. Convert the differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (\lambda x + x^3)y = 0$ into the standard Sturm-Liouville equation.

19. Examine whether the boundary value problem $x^2\frac{d^2y}{dx^2} + \lambda y = 0$; $y(1) = 0 = y(e)$ is a Sturm-Liouville problem or not?

20. Is the equation $(x^2 - y)dx + (y^2 - x)dy = 0$ an exact differential equation?

**Section C.** Answer any FOUR questions. Each question carries 10 marks.

21. (a) Find the possible eigenvalues and corresponding eigenfunctions for the Sturm-Liouville problem

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + (\lambda + 1)y = 0, \text{with}\, x, \in [0,\ 10],\ \lambda \in \mathbb{R},$$

such that, $y(0) = 0$, and $y'(10) = 0$. 5

(b) Show that the non zero eigenvalue of the Sturm-Liouville problem

$$\frac{d^2y}{dx^2} + \lambda y = 0, \text{with}\, x, \in [0,\ 1],\ \lambda \in \mathbb{R},$$

with $y(0) = 0$, and $y(1) = ky'(1) = 0$ are the solutions of the transcendental equation $\tan\sqrt{\lambda} = k\sqrt{\lambda}$, where $k$ is a real constant. 5

22. (a) Obtain the singular solution of the differential equation $y = px + \sqrt{1 + p^2}$. 4

(b) Solve the differential equation $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = 6x^5$. 6

23. (a) Solve the differential equation $\frac{d^2y}{dx^2} + a^2y = \sec ax$. 7

(b) Examine whether the equation $ydx + (x + e^y)dy = 0$ is exact or not? If exact then find its solution. 3

24. (a) Solve the differential equation $xy - \frac{dy}{dx} = y^3e^{-x^2}$. 5

(b) Find the particular integral of $(D^2+4)y = \sin 2x$. 5

25. (a) Find the orthogonal trajectories of the system of curves $r^2 = a^2 \cos 2\theta$. 5

(b) Find the integrating factor of the differential equation $(xy^2 - e^{x^{-3}})dx - x^2y\,dy = 0$ and solve it. 5

26. (a) Obtain the differential equation of all circles each of which touches the axis of x at the origin. 4

(b) Solve the differential equation $(1+e^{\frac{x}{y}})dx + e^{\frac{x}{y}}(1-\frac{x}{y})dy = 0$. 6

——— End ———

## Odd (Autumn) Semester Examination -2025

**Course Code:** PHYUGSEC2303
**Course Title:** Data analysis and Scientific writing
**Department of Appearing Students:** PHYSICS, SEM-III, 2nd Year
**Full Marks:** 60 **Time:** 2:30 hrs

**Answer question no.1 and any four from others**

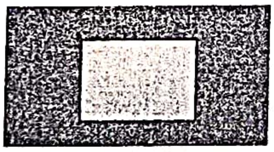
1. i) Write down the LaTeX command to generate the following table:

| Col 1 | Col 2 | Col 3 | Col 4 |
|---|---|---|---|
| 2 | 3 | 1 | 6 |
| 12 | 65 | -50 | 10 |
| -45 | 10 | 11 | 76 |
| 90 | 7.99 | -8.65 | 2.4 |

ii) Write down the LaTeX command to generate the equation: $x^2 + y^2 = 4\rho^2$

iii) Write down the LaTeX command to insert a figure stored in the file named "YOURROLLNUMBER.JPG

iv) Write the steps to be followed to insert the following shape in a Libre Office document.



v) Write the LaTeX command to produce the mathematical equation: $\mathrm{I(x)} = \int \mathrm{x}^2 \sin^3 \mathrm{x} d\mathrm{x}$

vi) Write down the LaTeX commands to prepare an article containing following components: Article title, Author name, Author address, Introduction, Method, Results and Discussion, Conclusion, References.

3+2+2+3+2+8

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2. Define data analysis and explain its main purpose. Give examples of structured and unstructured data. What are the key differences between qualitative and quantitative data?

3+3+4

3. Define primary and secondary data. Provide examples of each. What are the key advantages of observational studies? List three common methods of data collection and briefly describe them.

2+3+5

4. Define mean, median, and mode with examples. What is the significance of standard deviation in data analysis?

5+5

5. Explain the least squares method of curve fitting. Differentiate between a research paper and a review article.

5+5

6. Discuss the importance of ethics in scientific writing. What are the consequences of plagiarism in scientific research?

5+5

7. What are the main components of a research paper? Explain the role of review articles in advancing scientific knowledge.

5+5

# Autumn 2024
# End Semester Examination
## UCCUGAEC2302: Communicative English

**Full Marks: 80** **Time: 3 Hours**

## Section I

1. Answer **ANY TEN**of the following questions: **2x10=20 Marks**

a) What is *lingua franca*?
b) What is semantic gap?
c) What is passive listening?
d) What is skimming in reading?
e) What is a modal verb? Give an example.
f) Give an example of Grapevine communication.
g) What is active listening?
h) How many consonant sounds are there in the English language?
i) What is homophone? Give an example.
j) What is phrase? Give an example.
k) What are the channels of communication?
l) What is monologue?

m) What are diphthongs? Give an example.

n) What is intensive listening?

## Section II

2. Answer **ANY FOUR**questions: (word limit 200) **5x4=20 Marks**
a)Distinguish between formal and informal communication.

b) Define body language. Mention some of the gestures that we use to communicate.

c) Write a short dialogue between two friends discussing the recent price hike.

d) What is the difference between summary and paraphrase?

e) What are the important features of close reading?

f) What is business communication?

## Section III

3. Answer **ANY FOUR** of the following questions. (word limit 400) **10X4=40 Marks**

a) Identify the notable barriers to Effective Communication. Discuss the means to overcome those barriers. (6+4)

b) Attaching a Resume, write a job application for the post of a teacher in a college. (10)

c) Define and discuss the differences between Verbal Communication and Non-verbal Communication. (10)

d) Explain the different flows of communication in business organisations. (10)

e) Define Intra-personal Communication and Inter-personal Communication. Discuss the differences between Intra-personal Communication and Inter-personal Communication. (5+5)

f) Write an email to the Head of your Department requesting to extend the date of the submission of the tutorial project due to your illness. (10)

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## Aliah University

### Department of Economics

### Autumn (Odd) Semester Examination, January 2024

### For the 2nd Year 3rd semester students

**Paper Name-Development Studies** **Paper Code-ECOUGMDC2303**

**Full Marks-60** **Time- $2\frac{1}{2}$ Hours**

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## Group-A

**Answer any five** **(5x2 =10)**

1. What do you mean by sustainable development?
2. Differentiate between intermediate good and final good.
3. How can you find $NNP_{FC}$ from $GDP_{MP}$?
4. What is demographic dividend?
5. Differentiate between real GDP and Nominal GDP.
6. What is Okun's law
7. What is a merit good?

## Group B

**Answer any four** **(4x5=20)**

8. What is tax? Differentiate between direct tax and indirect tax. (2+3)
9. Differentiate between economic growth and economic development.
10. What are the negative impacts of high unemployment on an economy?
11. Why do you think that basic school education and health should be provided at almost free of cost by the government?
12. Explain the viscous cycle of poverty
13. Differentiate between:

a. Voluntary Unemployment and Involuntary unemployment

b. Absolute Poverty and Relative Poverty

## Group C

**Answer any three** **(3x 10 =30)**

14. Briefly discuss the Sustainable development Goals of the United Nation. (10)
15. How does the following lead to unemployment in Indian economy?
    a. Presence of labour union.
    b. Efficiency wage (5+5)
16. Do you think national income or GDP is a good indicator of welfare for an economy?
17. Explain positive and negative externalities in case of both consumption and production.
18. A. What are the causes of demand pull and cost push inflation?
    B. How does the RBI control inflation? (6+4)